Boolean Algebra as the Geometry of Clopen Sets

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The subsets of a finite set can be combined by union, intersection and complement. Nothing in that calculation suggests a new topology until we ask what a Boolean algebra looks like when the underlying set is no longer given. Stone's idea is to regard homomorphisms into the two-element algebra as points. The algebra then manufactures its own compact topological space, and every Boolean operation becomes an operation on clopen sets.[1]

§1.1 A Boolean ring is a calculus of exclusive choices

For a set X, use symmetric difference and intersection as operations on $\mathcal{P}(X)$:

$$A + B = (A \setminus B) \cup (B \setminus A), \qquad AB = A \cap B.$$

Every element is idempotent under multiplication:

$$A^2 = A.$$

Indeed, $\mathcal{P}(X)$ becomes a commutative ring with additive identity $\varnothing$, multiplicative identity X, and $A + A = \varnothing$. Characteristic functions identify it with the ring of functions $X \to \mathbb{F}_2$, with pointwise addition and multiplication.

Conversely, suppose R is a unital ring satisfying $a^2 = a$ for every $a \in R$. Expanding $(a+a)^2 = a + a$ and using $a^2 = a$ shows $2a = 0$. Expanding $(a+b)^2 = a + b$ then shows $ab + ba = 0$, so $ab = ba$ in characteristic two. Thus every Boolean ring is commutative and has characteristic two (unless it is the zero ring).

The Boolean-algebra operations can be recovered from the ring:

$$a \wedge b = ab, \qquad \neg a = 1 + a, \qquad a \vee b = a + b + ab.$$

For subsets these become intersection, complement and union. The ring formalism is useful because algebraic homomorphisms already know how to preserve all three logical operations.

§1.2 Points appear as consistent truth assignments

Let B be a Boolean algebra with $0 \neq 1$. A map

$$\chi : B \longrightarrow \{0, 1\}$$

preserving $0, 1, \wedge, \vee, \neg$ is a complete Boolean truth assignment. Its true elements form

$$U_\chi = \{b \in B : \chi(b) = 1\}.$$

This is an *ultrafilter*: it contains 1 but not 0, is closed under finite meets, is upward closed, and for each $b \in B$ contains exactly one of b and $\neg b$.

[1] The algebraic and topological constructions follow Davey and Priestley, *Introduction to Lattices and Order*, and P. T. Johnstone, *Stone Spaces*.

Conversely, given such an ultrafilter U, put $\chi_U(b) = 1$ if $b \in U$, and 0 otherwise. The ultrafilter axioms verify the Boolean homomorphism laws. Hence

$$\boxed{\{\text{ultrafilters on } B\} \cong \mathrm{Hom}_{\mathrm{Bool}}(B, 2).}$$

The two descriptions are interchangeable: an ultrafilter is a consistent way to declare each Boolean proposition true or false.

For a finite set X, every ultrafilter on $\mathcal{P}(X)$ comes from exactly one point $x \in X$:

$$U_x = \{A \subseteq X : x \in A\}.$$

Indeed, since X is the union of its finitely many singleton subsets, the ultrafilter property forces one singleton to be true; two different singletons cannot both be true because their intersection is empty. A finite Boolean algebra can therefore recover its underlying points without knowing them in advance.

§1.3 An infinite example produces an extra point

Let B be the Boolean algebra of all *finite or cofinite* subsets of $\mathbb{N}$. For each $n \in \mathbb{N}$ we have the principal ultrafilter

$$U_n = \{A \in B : n \in A\}.$$

There is also a different ultrafilter:

$$U_\infty = \{A \in B : \mathbb{N} \setminus A \text{ is finite}\}.$$

It contains every cofinite set, no finite set, and decides every element of B, so it is an ultrafilter.

These are all the ultrafilters. If an ultrafilter contains a finite set F, the finite-union property forces it to contain a singleton $\{n\} \subseteq F$, hence it equals U_n. Otherwise it contains no finite set and therefore must contain the complement of every finite set, making it U_∞.

The ultrafilters have thus produced the space

$$S(B) = \mathbb{N} \cup \{\infty\}.$$

The extra point was not inserted artificially: it is the truth assignment that accepts all cofinite sets and rejects every finite set.

§1.4 The topology is forced by the algebra

For each $b \in B$, define

$$\widehat{b} = \{U \in S(B) : b \in U\}.$$

The ultrafilter laws immediately give

$$\widehat{a \wedge b} = \widehat{a} \cap \widehat{b}, \qquad \widehat{a \vee b} = \widehat{a} \cup \widehat{b}, \qquad \widehat{\neg a} = S(B) \setminus \widehat{a}.$$

These sets are closed under finite unions and complements; declaring them open makes them automatically *clopen*. They form a basis of a topology on $S(B)$.

For our finite–cofinite algebra,

$$\widehat{\{n\}} = \{U_n\},$$

so each ordinary point n is isolated. For a cofinite subset $\mathbb{N}\setminus F$, the corresponding clopen set is

$$\widehat{\mathbb{N}\setminus F} = \{U_\infty\} \cup \{U_n : n \notin F\}.$$

Every neighborhood of U_∞ therefore contains all but finitely many natural-number points. This is exactly the *one-point compactification* of the discrete space $\mathbb{N}$.

Compactness can be checked directly. An open cover has some member containing U_∞, which already covers all but finitely many points. Finitely many additional members cover those remaining points. The topology was constructed from Boolean operations alone, yet it has recovered a classical compact space.

§1.5 Stone's theorem and the role of ultrafilter extension

Stone's representation theorem states that every Boolean algebra B is isomorphic to the Boolean algebra $\mathrm{Clop}(S(B))$ of clopen subsets of its ultrafilter space:

$$\boxed{B \xrightarrow{\sim} \mathrm{Clop}(S(B)), \qquad b \longmapsto \widehat{b}.}$$

We have already verified that the map preserves Boolean operations. The remaining issues are injectivity and surjectivity, which reveal where compactness enters the theory.

If $a \neq b$, their symmetric difference is nonzero. By the Boolean prime ideal theorem, a nonzero element belongs to some ultrafilter. Such an ultrafilter distinguishes a and b, so $\widehat{a} \neq \widehat{b}$. This proves injectivity.

For surjectivity, every clopen $C \subseteq S(B)$ is open, hence covered by basic sets $\widehat{b_i}$ contained in C. The ultrafilter space is compact, so finitely many of them suffice:

$$C = \widehat{b_1} \cup \cdots \cup \widehat{b_m} = \widehat{b_1 \vee \cdots \vee b_m}.$$

Thus every clopen subset comes from an element of B.

The general compactness of $S(B)$ and the ultrafilter-extension step can be obtained from the Boolean prime ideal theorem, a choice principle weaker than the full axiom of choice. In the finite–cofinite example we needed neither an abstract extension lemma nor an appeal to an existence theorem: every point and every basic open set could be listed explicitly.

§1.6 Why the correspondence reverses arrows

Suppose $h : B \to C$ is a Boolean-algebra homomorphism. A point of $S(C)$ is an ultrafilter $U \subseteq C$. Pull it back:

$$h^{-1}(U) = \{b \in B : h(b) \in U\}.$$

It is an ultrafilter of B, so h induces a continuous map

$$S(h) : S(C) \longrightarrow S(B), \qquad U \longmapsto h^{-1}(U).$$

Continuity follows from the actual calculation

$$S(h)^{-1}(\widehat{b}) = \widehat{h(b)}.$$

Notice the direction reversal: an algebra homomorphism $B \to C$ gives a map of spaces $S(C) \to S(B)$, not $S(B) \to S(C)$.

Conversely, if $f : X \to Y$ is a continuous map between compact Hausdorff totally disconnected spaces, inverse image sends clopen sets of Y to clopen sets of X:

$$f^{-1} : \mathrm{Clop}(Y) \longrightarrow \mathrm{Clop}(X).$$

That operation also reverses arrows. Stone duality states that these two contravariant constructions give an equivalence between Boolean algebras and compact Hausdorff zero-dimensional spaces.

This is the deeper reason Boolean algebra belongs simultaneously to logic and topology. The algebra describes which yes/no statements can be combined; the points are consistent evaluations of those statements; and continuity means that truth regions pull back to truth regions. If we replace clopen sets by all open sets, complements need not remain open, and the resulting Heyting algebra may fail excluded middle. Classical Boolean logic corresponds to the geometry of clopen predicates.